Statistics and Trends: What Shapes Student's Math's Grade

Pujitha Bheemana | 25101281

Introduction:

I wanted to find out what really drives a student's final mark in maths. The data covers 395 pupils from two Portuguese secondary schools, and I look at three things in turn: how the final grades are spread out, how closely they follow the marks students already had earlier in the year, and whether family background, past failures or social life explain anything extra. A short critical note at the end covers what the numbers can and cannot tell us.

About Dataset:

Each of the 395 rows is one pupil, described by 33 variables. These include three grades out of 20 (G1, G2 and the final G3), past class failures, weekly study time, parents' education, absences, and self-rated social life and health. Nothing was missing and there were no duplicates, so I used every record.

Where the Data Comes From:

I used the public *Student Performance* dataset from the UCI Machine Learning Repository, which was gathered for a study by Cortez, P. and Silva, A. (2008), Using Data Mining to Predict Secondary School Student Performance, *Proceedings of FUBUTEC 2008*, pp. 5-12, Porto.

1. Histogram of final grades:

Figure 1 shows that most final grades land around the middle of the scale. The average is 10.42 and the median is 11, with a standard deviation of 4.58. A skewness of -0.73 tells us the tail stretches further to the left. About two thirds of pupils (64%) scored between 8 and 14, and the single most common grade is 10, earned by 56 students. What stands out is the lone bar at 0: 38 pupils (9.6%) got no marks at all, yet nobody got 1, 2 or 3. That gap suggests they are a different group rather than the bottom of a smooth curve. Leave them out and the mean climbs to 11.52 while the standard deviation falls to 3.23, so these zeros account for a large share of the spread. Every one of them also has zero absences, which makes me think they dropped out or missed the final assessment instead of attending and failing.

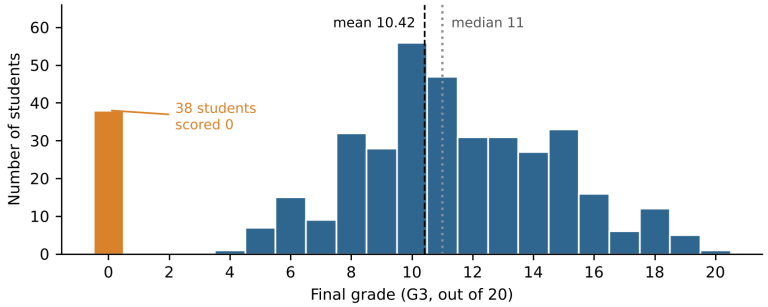


Figure 1. Distribution of final grades (G3). Orange marks students with a grade of zero.

2. Scatter plot of G2 and G3:

Figure 2 compares the second-period grade (G2) with the final grade. The points follow a fairly straight line, with a correlation of 0.90, which means G2 explains about 82% of the variation in G3. The fitted line is G3 = 1.10 x G2 - 1.39. Most points sit just under the G3 = G2 line: on average pupils ended 0.3 marks below their G2 score, and only 28% improved. Around the line the scatter is roughly two marks either way in the middle of the range, and it narrows for high achievers. The row of points along G3 = 0 is the same group from Figure 1. Many of them had ordinary G2 scores between 5 and 10, which no straight line could have predicted. Dropping students with G2 = 0 barely matters (r = 0.89), so a few extreme cases are not driving the result.

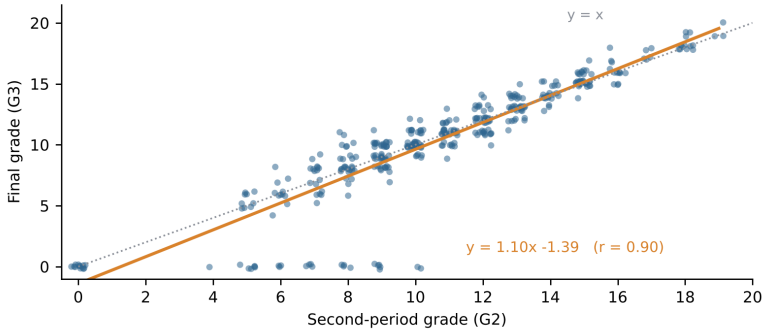


Figure 2. G2 against G3 (small jitter added to show overlapping points). Solid line: least-squares fit; dotted line: G3 = G2.

3. Heatmap of student performance factors:

The heatmap in Figure 3 compares the three grades with nine background and lifestyle variables. The grades sit together in a strong block (0.80 to 0.90), which is expected since they measure the same skill at different points in the year. Outside that block the links are modest. Past failures stand out most (r = -0.36): pupils with none average 11.25 (n = 312), while those with one, two or three failures average 8.12, 6.24 and 5.69. Mother's and father's education both lean positive with G3 (0.22 and 0.15) and are linked to each other (0.62), so they probably reflect a single household effect rather than two. Study time is weak (0.10), and absences, health and weekend drinking sit close to zero. One side finding is that going out and weekend drinking tend to rise together (0.42).

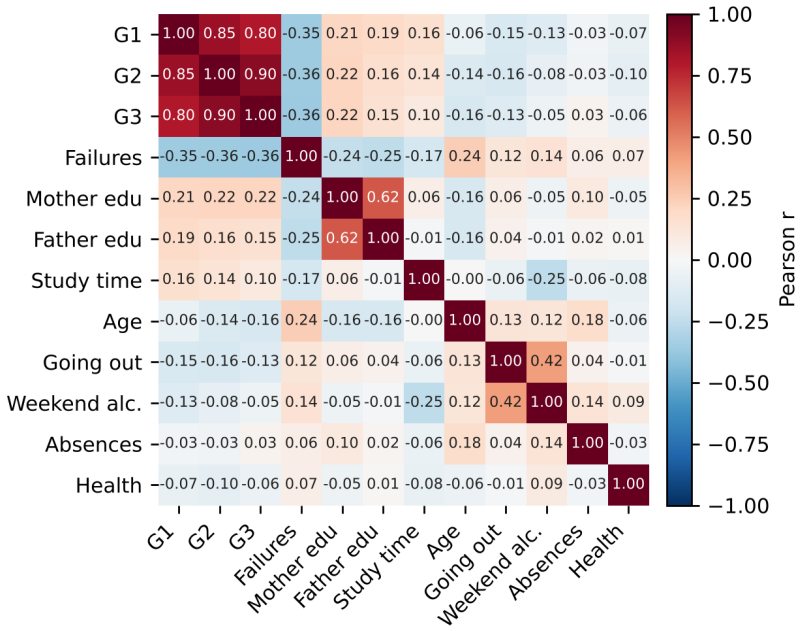


Figure 3. Correlation matrix for the three grades and selected background and lifestyle variables.

Conclusion:

So what shapes a student's maths grade? Mainly where they already stood. Second-period marks alone explain about 82% of the variation in final grades, and a record of failed classes is the next clearest signal: pupils with no failures average 11.25, against 5.69 for those with three. Parents' education adds a small boost, while study time, going out, health and drinking show weak or no relationship.

Three cautions are worth keeping in mind. First, earlier grades predict the final mark so well partly because they measure the same ability, so they forecast more than they explain. Second, the 38 zero-grade pupils, none with recorded absences, distort simple correlations: absences seem unrelated to grades overall (r = 0.03), yet pupils with some absences average 11.22 against 8.44 for those with none. Third, the data are observational and come from just two schools, so these are associations, not proof of cause. Next I would treat the zero-grade pupils as a separate outcome and run a regression that controls for earlier grades.

GIT-HUB link: https://github.com/Pujitha-0716/Statistics-and-Trends---Report